



IOT FRAMEWORKS AND ALLIANCES: The Focus is on Interoperability

The Internet of Things (IoT) is a trending topic of discussion and has created a buzz in computer circles. It is also a topic which is being subjected to intensive research. This article provides details of various platforms that enable the IoT and the efforts being made to develop standards that increase interoperability amongst devices.

The Internet of Things (IoT) refers to smart machine-to-machine (M2M) interconnection technology enabled by secure connectivity and appropriate infrastructure, to reliably transform data into useful information for people, businesses and institutions with the right analytics. The appropriate infrastructure could be hybrid compute and store mechanisms like the cloud stack or Big Data technologies.

Lots of IoT frameworks and platforms with different enterprise offerings and alliances have come up recently to increase the connectivity of devices in private and public networks. The key to a successful IoT implementation is the interoperability between various devices and a robust platform that enables this with a powerful analytics framework, which further allows applications to be built on top of it. Additionally, this platform also needs to provide value added services in order to attract enterprise adoption.

This article covers the various platforms that enable the IoT and the efforts being made to develop standards that increase the interoperability amongst devices.

Frameworks and platforms

The basic components of the IoT value chain highlighted in Figure 1 are part of the offerings that are provided by various players in this space. Due to the evolution of these platforms, the collective features offered vary, and over a period of time may converge into these architectural blocks.

IoT platforms have evolved from two different directions. First, those providing connectivity to devices in private networks moved to the Internet and the public cloud. The other approach involved the cloud providers adding IoT protocols and application support to enhance the reach of their platforms to multiple industries. Let's take a look at the different frameworks and platform providers that are prevalent in the industry.

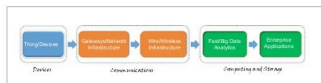


Figure 1: IoT value chain